



YEAR 7 MATHEMATICS · CHAPTER 10

Misleading graphs

Lesson 2 of 8

**In this lesson you will learn to:**

- spot features that make a graph misleading
- explain the false impression a misleading graph gives
- redraw a misleading graph so it is fair

Name

Class

Teacher

Before you start

You will need

Pencil

Ruler

Syllabus: MA4-DAT-C-01: classifies and displays data using a variety of graphical representations

WE DO

Grey boxes. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box.

- 1** Work through **Level 1**, then **Level 2**, then **Level 3**. Try the **extension** if you finish early.

- 2** At the end, **show off your skill**: answer one exit ticket question, tear it off and hand it in.

Lesson map

Page	What is on it	Done
3	Learning intentions, key terms, notes and Level 1 teacher examples	☐
4	Level 1: YOU DO practice questions	☐
5	Level 2: WE DO teacher examples and YOU DO practice questions	☐
6	Level 2 and 3: WE DO teacher examples and YOU DO practice questions	☐
7	Level 3: YOU DO practice questions	☐
8	Level 3: YOU DO practice questions	☐
9	Extension questions and the tear-off exit ticket	☐
10	Lesson summary: a worked example for every level	☐

YOU DO

On your own: Warm-up. These skills will help you in this lesson. Show your working.

1 $64 - 58$

3 2×2

2 $6 \div 2$

4 What should every graph have at the top?



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 10.02

Misleading graphs

Date _____

Learning intentions

We are learning to:

- spot features that make a graph misleading
- explain the false impression a misleading graph gives
- redraw a misleading graph so it is fair

Success criteria

I can:

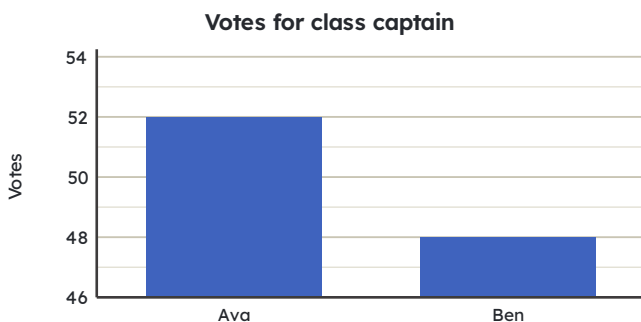
- ☐ check whether the vertical axis starts at 0 and has an even scale
- ☐ explain how a graph makes a difference look bigger or smaller than it really is
- ☐ redraw a graph with a fair scale, title and labels

Key terms Write the definition as your teacher explains it.

Misleading graph
Scale
Axis

Notes

Level 1 · Getting started


WE DO

With your teacher: copy what they write on the board.

Example 1

What is misleading about this graph?

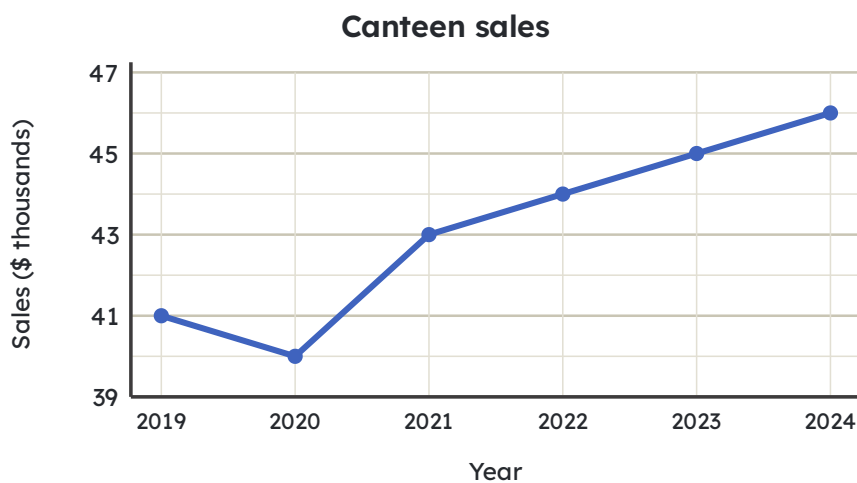
Example 2

Ava's bar looks 3 times as tall as Ben's. How many more votes did Ava really get?

YOU DO On your own: Set A. Is each feature fair or misleading? Write F or M.

- | | |
|---|--|
| 1 The vertical axis starts at 0. <input type="text"/> | 6 The graph has a clear title. <input type="text"/> |
| 2 The vertical axis starts at 90, not 0. <input type="text"/> | 7 Some years are left out of a line graph. <input type="text"/> |
| 3 The scale goes 0, 10, 20, 50, 100 with equal gaps. <input type="text"/> | 8 A tilted 3D pie makes the front sector look bigger. <input type="text"/> |
| 4 Both axes are labelled with units. <input type="text"/> | 9 The gridlines are evenly spaced. <input type="text"/> |
| 5 One bar is drawn wider than the others. <input type="text"/> | 10 A picture is drawn taller and wider to show a bigger value. <input type="text"/> |

YOU DO On your own: Set B. Use the canteen sales line graph.

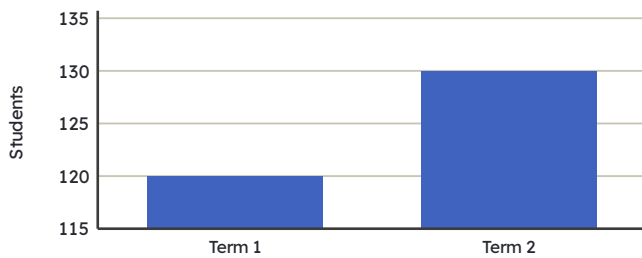


- | | |
|---|--|
| 1 What were sales in 2019? <input type="text"/> | 6 On the graph, 2024 looks how many times higher than 2020? <input type="text"/> |
| 2 In which year were sales lowest? <input type="text"/> | 7 True or false: sales more than doubled. <input type="text"/> |
| 3 What does the vertical axis start at? <input type="text"/> | 8 Which year had the biggest rise? <input type="text"/> |
| 4 Should it start at 0? (yes/no) <input type="text"/> | 9 Are the axes labelled? (yes/no) <input type="text"/> |
| 5 How much did sales rise from 2020 to 2024? <input type="text"/> | 10 Name one thing that makes it misleading. <input type="text"/> |

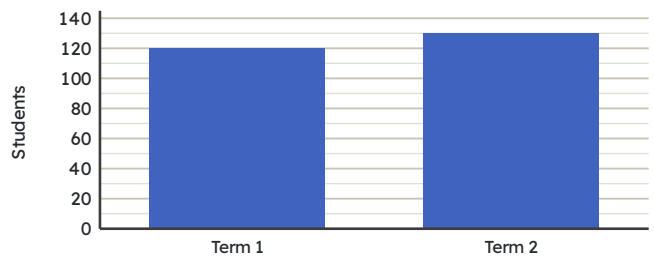
Level 2 · Building up

Graphs P and Q show the same data: 120 students bought lunch in Term 1 and 130 in Term 2.

Graph P: students buying lunch



Graph Q: students buying lunch



WE DO With your teacher: copy what they write on the board.

Example 3 Which graph makes the increase look bigger? Explain why.

Example 4 In Graph P, Term 2's bar looks 3 times as tall. Is that true? How many more students were there?

YOU DO On your own: Set C. Questions 1–5 use Graphs P and Q above.

1 Graph P: what does the vertical axis start at?

2 Graph Q: what does the vertical axis start at?

3 How many more students bought lunch in Term 2?

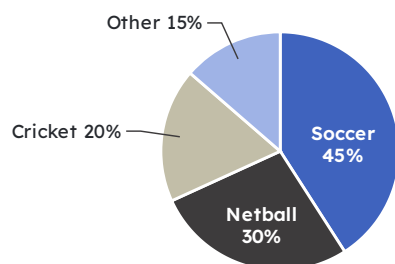
4 In Graph P, Term 2's bar looks how many times as tall as Term 1's?

5 Which graph would the canteen use to show a big jump? Why?

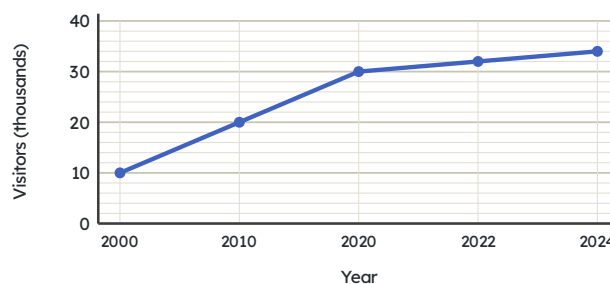
YOU DO

On your own: Set C continued. Questions 6–7 use the sector graph and 8–10 use the line graph.

Favourite sport of Year 7



Visitors to the town museum



6

Sector graph: add the percentages. What is the total?

9

Line graph: how many years are between 2020 and 2022?

7

Why is the sector graph misleading?

10

Why is the line graph's horizontal axis misleading?

8

Line graph: how many years are between 2000 and 2010?



Level 3 · Challenge

WE DO

With your teacher: copy what they write on the board.

Example 5

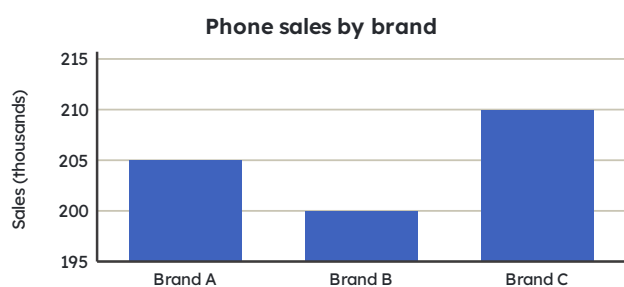
Redraw the class captain graph (page 3) fairly: Ava 52 votes, Ben 48 votes.

Example 6

Who might want to use the misleading class captain graph, and why?

YOU DO

On your own: Set D. Questions 1–3 use the phone sales graph.



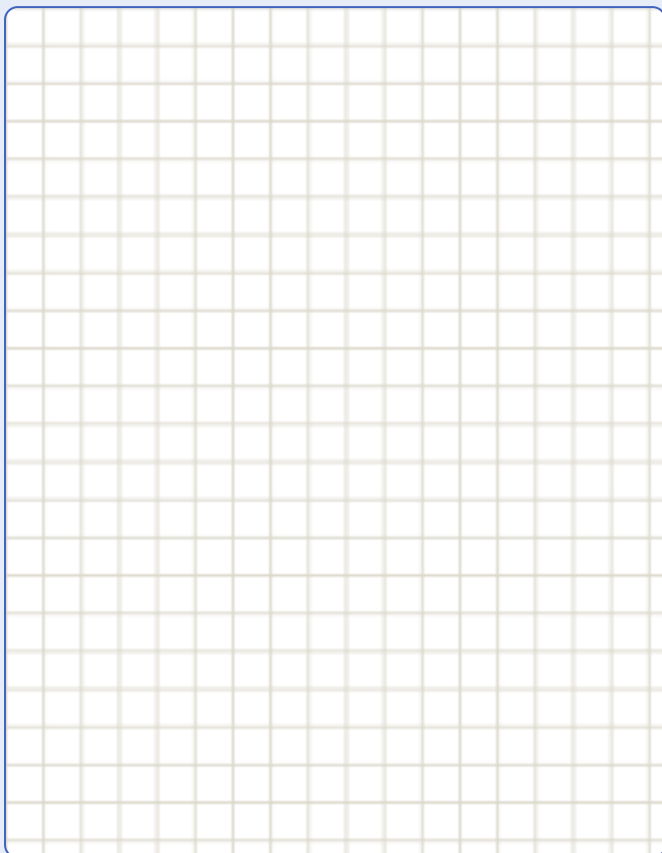
- 1 What does the graph suggest about Brand C compared with Brand B?

- 2 What are the real sales of Brands B and C? What is the difference?

- 3 Redraw the phone sales graph fairly. Start the vertical axis at 0, add a title and label both axes.

YOU DO On your own: Set D continued.

- 4 Redraw the canteen sales graph (Set B) with the vertical axis starting at 0. Sales (\$ thousands) for 2019 to 2024: 41, 40, 43, 44, 45, 46.



- 5 How does your new canteen graph change the impression?

- 6 A headline says “Crime has doubled!” The graph’s axis starts at 50 and crimes rose from 52 to 54. Is the headline true? Explain.

- 7 A house is drawn twice as tall **and** twice as wide to show prices doubling. Why is this misleading?

- 9 List three things every fair graph should have.

- 8 Give two reasons someone might make a misleading graph on purpose.

- 10 A graph shows only the months when sales went up. Why is leaving data out misleading?

Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

A shop's sales doubled. On a poster, this year's circle has **twice the width and twice the height** of last year's. How many times bigger does it look?

1. Width $\times 2$ and height $\times 2$
2. The area (how big it looks) is $2 \times 2 = 4$ times as big
3. Sales only doubled, so the picture exaggerates



last year



this year

Answer: it looks 4 times bigger

YOU DO

On your own: read the worked example, then try these.

- E1** A picture is made 3 times as wide and 3 times as tall. How many times bigger does it look?

- E2** Sales went from 100 to 150. A bag is drawn 1.5 times as tall and 1.5 times as wide. How many times bigger does it look? How should it be drawn fairly?

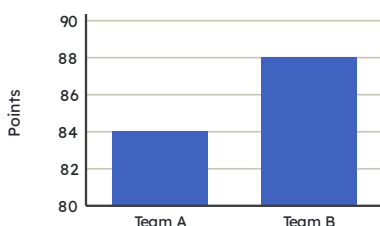
✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Points scored



Level 1

What number does the vertical axis start at?

Level 2

Team B's bar looks twice as tall. How many more points did B really score?

Level 3

Why is this graph misleading? How would you fix it?

Lesson summary

Missed the lesson? Start here.

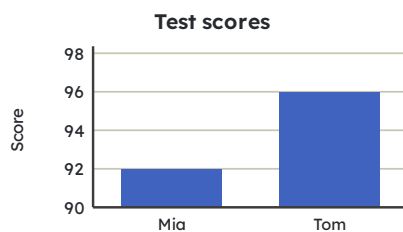
Is this graph fair?

1. Does the vertical axis start at 0 (or show a break)?
2. Is the **scale even** (equal gaps = equal amounts)?
3. Are the **title and axis labels** there?
4. Is **all the data** shown, with bars the same width?

Level 1

A column graph's vertical axis starts at 90. What is the problem?

1. Bar heights are measured from 90, not from 0
2. So small differences look huge

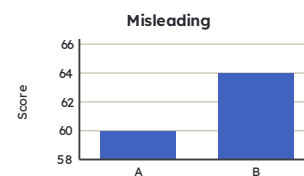


Answer: it is misleading. Start the axis at 0.

Level 2

Scores of 60 and 64 on an axis starting at 58. How many times as tall does the 64 bar look? What is the real difference?

1. Heights on the graph: $60 - 58 = 2$ and $64 - 58 = 6$
2. $6 \div 2 = 3$, so it looks 3 times as tall
3. Real difference: $64 - 60 = 4$

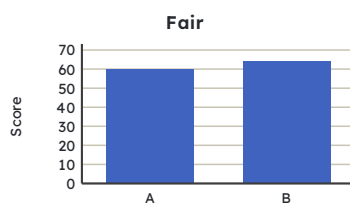


Answer: looks 3 times as tall; really only 4 more

Level 3

Redraw the Level 2 graph fairly.

1. Start the vertical axis at 0 with an even scale
2. Plot the same values: 60 and 64
3. Add a title and label both axes



Answer: the bars now look almost the same, which is the truth

Extension

A picture is drawn twice as wide and twice as tall. How many times bigger does it look?

1. Area grows by width $\times$ height
2. $2 \times 2 = 4$



Answer: 4 times bigger, though the value only doubled

This space is left blank on purpose. It is the back of your exit ticket.